

Jeffrey Utley

PhD Candidate, Mathematics — Computational Science

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U.S. Citizen

RESEARCH SUMMARY

PhD candidate at Purdue University developing statistically validated synthetic data generation algorithms and reproducible scientific software for turbulence-driven imaging and simulation. Three years of DoD-funded research in active collaboration with AFRL and AFIT, with first-author publications in *Journal of the Optical Society of America A*, *Optical Engineering*, and SPIE proceedings, and pip-installable Python packages validated on measured datasets. Targeting Research Scientist and R&D roles at national laboratories and in industry, with focus on modeling, estimation, and simulation; computational imaging; signal processing; and scientific software.

TECHNICAL SKILLS

Methods: algorithm design and analysis; synthetic data generation; inverse problems and parameter estimation; spectral analysis (FFT, PSD, PCA); numerical optimization (first-order methods, Mirror Descent); statistical modeling (Gaussian models, Poisson point processes); Bayesian inference; signal and image processing

Optics & Imaging: aero-optics; optical turbulence characterization; atmospheric optics; phase screen generation; wavefront synthesis; boundary layer turbulence; Fourier optics; computational imaging; electro-optical imaging

Programming: Python, C/C++, MATLAB, Julia; object-oriented design and modular package architecture

Scientific Computing: NumPy, SciPy, pandas, Matplotlib, Jupyter, OpenCV, seaborn

Machine Learning: PyTorch, JAX

HPC & Systems: Linux/UNIX, bash, SSH, shell scripts, Slurm (Purdue clusters)

Reproducibility & Tools: Git/GitHub, pip/conda, setuptools/pyproject, Make; reproducible builds, automated testing (pytest), and technical documentation

EDUCATION

Ph.D., Mathematics — Purdue University

Expected Dec 2026

Concentration: Computational Science • GPA: 4.00 • Advisors: Prof. Gregory Buzzard and Prof. Charles Bouman

Dissertation: Synthetic Data Generation for Aero-Optics

Selected Coursework: Foundations of Computational Imaging; Digital Image Processing; Digital Cameras & Fourier Optics; Deep Learning; Numerical Optimization; Numerical Linear Algebra; Numerical Analysis; Probability

B.S., Mathematics (Honors) — University of Tennessee, Knoxville

May 2022

Minor: Computer Science • GPA: 3.98

RESEARCH EXPERIENCE

Graduate Research Assistant — Purdue University, Department of Mathematics

Jan 2023 – Present

Academic-year appointment; paused each summer for the research placements below.

- Designed **ReVAR**, a novel data-driven algorithm producing synthetic data that matches measured statistics from physical experiments with worst-case **4% NRMSE** on the temporal power spectrum — a substantial improvement over existing methods.
- Developed **BoilingFlow**, extending a Fourier-based data synthesis method with novel automated parameter estimation and a spatial-anisotropy extension; validated on measured and simulated datasets (worst-case 12% NRMSE on the temporal power spectrum).
- Built statistical estimation pipelines (FFT/PSD, PCA, Gaussian models) over measured phase screen data spanning >10,000 time-samples per dataset.

- Delivered all results as modular, version-controlled, **pip-installable Python packages** with reproducible examples and full documentation; ran large-scale parameter sweeps on Purdue HPC clusters via Slurm; produced first-author publications in *Journal of the Optical Society of America A*, *Optical Engineering*, and SPIE proceedings.
- Collaborate weekly with DoD program scientists to translate experimental requirements into validated software deliverables.
- Authored four first-author papers (two journal, two conference) in <4 years; presented results through eight conference talks at national conferences.

Summer Research Placements

AFRL Scholars Program (USRA) — Air Force Research Laboratory (virtual)

May 2026 – Sept 2026

Mentor: Jeremy Vorenberg

- Second consecutive AFRL Scholars selection; continues the Purdue–AFRL/AFIT collaboration on synthetic data generation for turbulence-driven imaging and simulation with Prof. Matthew Kemnetz.

Doctoral Research Fellow (ORISE) — Air Force Institute of Technology (virtual)

May 2025 – Aug 2025

Mentor: Prof. Matthew Kemnetz

- Held the lead computational-science role on the Purdue–AFRL/AFIT collaboration; advanced the BoilingFlow algorithm, delivered weekly virtual technical briefings to AFIT faculty and contributed to a SPIE conference paper deliverable.

AFRL Scholars Program (USRA) — Air Force Research Laboratory

May 2024 – Aug 2024

Mentor: Prof. Matthew Kemnetz

- Lead computational-science contributor on the Purdue–AFRL/AFIT collaboration and first AFRL Scholars selection; developed and evaluated the ReVAR algorithm against measured wind tunnel data; delivered an end-of-summer technical briefing to AFRL staff and contributed to a SPIE conference paper.

RESEARCH SOFTWARE

AOModel — pip-installable Python package implementing the ReVAR algorithm for data-driven synthesis of optical turbulence data. Validated on two wind tunnel datasets with worst-case 4% NRMSE on the temporal power spectrum. Includes thorough documentation and reproducible examples. github.com/jeffreyutley/aomodel_public

BoilingFlow — pip-installable Python package implementing the BoilingFlow algorithm, extending a Fourier-based data synthesis method with automated parameter estimation and a spatial-anisotropy extension. Validated on two wind tunnel datasets (worst-case 12% NRMSE on the temporal power spectrum). Includes thorough documentation and reproducible examples. github.com/jeffreyutley/boiling_flow

PUBLICATIONS

Refereed Journal Articles

Utley, J., Buzzard, G., Bouman, C., & Kemnetz, M. Boiling flow estimation for aero-optic phase screen generation. *Optical Engineering* (in press). Preprint: [arXiv:2601.12171](https://arxiv.org/abs/2601.12171).

Utley, J., Buzzard, G., Bouman, C., & Kemnetz, M. (2026). ReVAR: A data-driven algorithm for generating aero-optic phase screens. *Journal of the Optical Society of America A*, 43(7), C30-C44. DOI: [10.1364/JOSAA.600450](https://doi.org/10.1364/JOSAA.600450). Preprint: [arXiv:2604.02326](https://arxiv.org/abs/2604.02326).

Lind, J. & **Utley, J.** (2022). Phase transition for a family of complex-driven Loewner hulls. *Involve, a Journal of Mathematics*, 15(3), 447–474. DOI: [10.2140/involve.2022.15.447](https://doi.org/10.2140/involve.2022.15.447). Preprint: [arxiv:2106.14940](https://arxiv.org/abs/2106.14940).

Conference Proceedings

Utley, J., Buzzard, G., Bouman, C., & Kemnetz, M. (2025). Boiling flow parameter estimation from boundary layer data. *SPIE Optics + Photonics*. DOI: [10.1117/12.3063655](https://doi.org/10.1117/12.3063655). Preprint: [arXiv:2602.10394](https://arxiv.org/abs/2602.10394).

Utley, J., Buzzard, G., Bouman, C., & Kemnetz, M. (2024). Data-driven synthetic wavefront generation for boundary layer data. *SPIE Optics + Photonics*. DOI: [10.1117/12.3027740](https://doi.org/10.1117/12.3027740). Preprint: [arXiv:2409.04873](https://arxiv.org/abs/2409.04873).

CONFERENCE PRESENTATIONS

Directed Energy S&T Symposium 2026 — “ReVAR: A data-driven algorithm for generating aero-optic phase screens.”

Directed Energy Education Workshop 2026 — “ReVAR: A data-driven algorithm for generating aero-optic phase screens.”

Electronic Imaging 2026 — “ReVAR: A data-driven algorithm for generating aero-optic phase screens.”

SPIE Optics + Photonics 2025 — “Boiling flow parameter estimation from boundary layer data.”

Directed Energy S&T Symposium 2025 — “Data-driven synthetic wavefront generation for boundary layer data.”

Electronic Imaging 2025 — “Synthetic wavefront generation for aero-induced turbulence using boundary layer data.”

SPIE Optics + Photonics 2024 — “Data-driven synthetic wavefront generation for boundary layer data.”

Directed Energy S&T Symposium 2024 — “Synthetic wavefront generation for aero-optics correction.”

HONORS, LEADERSHIP, SERVICE, & TEACHING

Fellowships: Doctoral Research Fellow, Oak Ridge Institute for Science and Education / AFIT (2025)

Awards: SPIE Student Conference Support Award (2024); Purdue College of Science Travel Award (2024); John H. Barrett Memorial Prize, University of Tennessee (2022).

Leadership: President, SIAM Student Chapter, Purdue University (2026–Present); Co-organizer, CCAM Lunch Seminar, Purdue Mathematics (2025–Present).

Service: Reviewer, *Optical Engineering* (SPIE), 2025–Present (2 manuscripts).

Teaching: Graduate TA, Purdue University (2022); Instructor, Wolverine Pathways at University of Michigan — math instruction for under-resourced middle and high school students (2022–2023); Undergraduate TA & Math Tutor, University of Tennessee (2020–2022).